

Diffusion models

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Building AI/ML applications
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Four main tasks in computer vision

Regression and
classification of images



Cat? Yes

Dog? Yes

Bat? No

Feature extraction



Cat color: black

Dog color: brown

Cat's eye color:
green

Dog's eye color:
brown

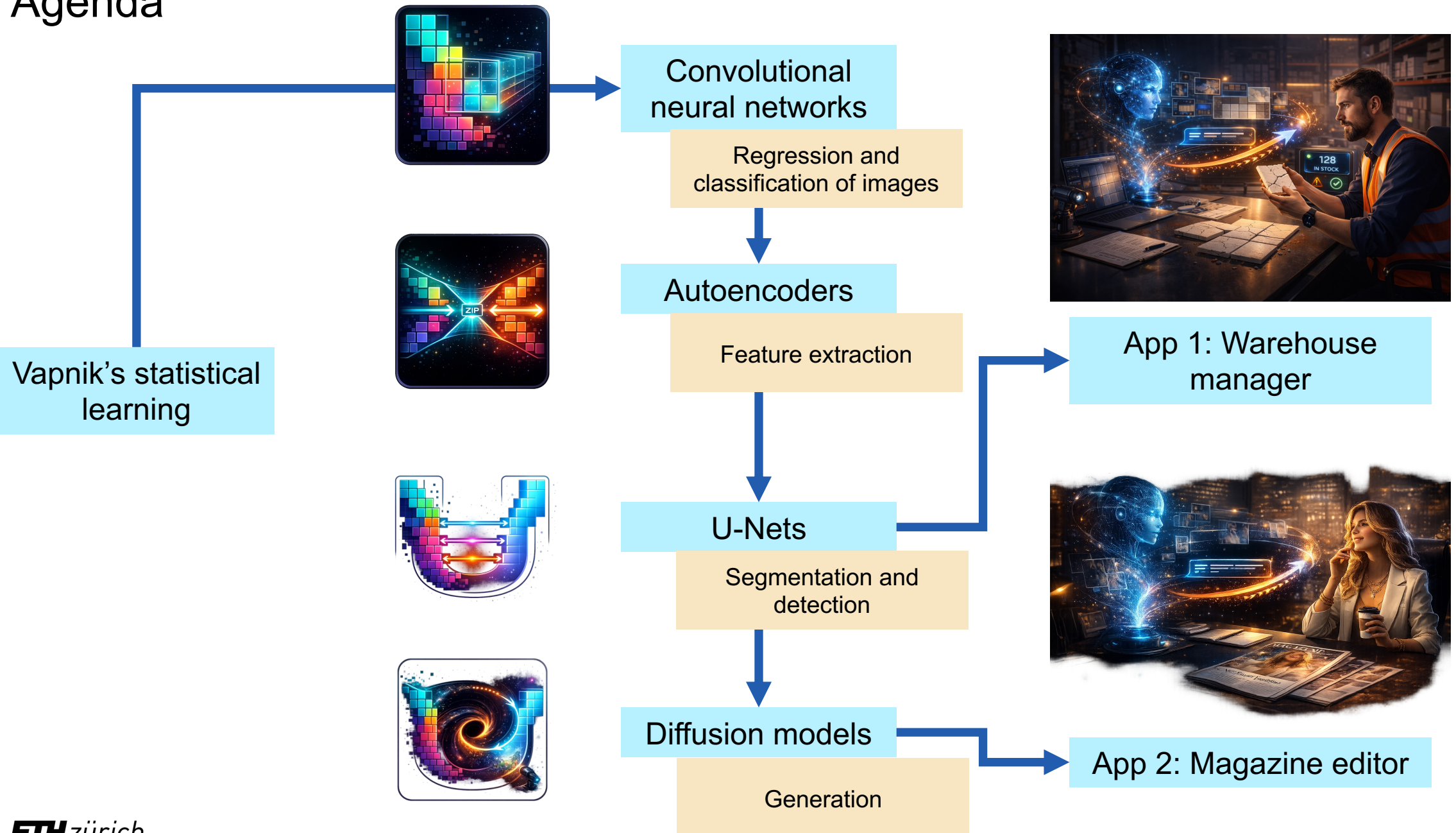
Segmentation and
detection



Generation



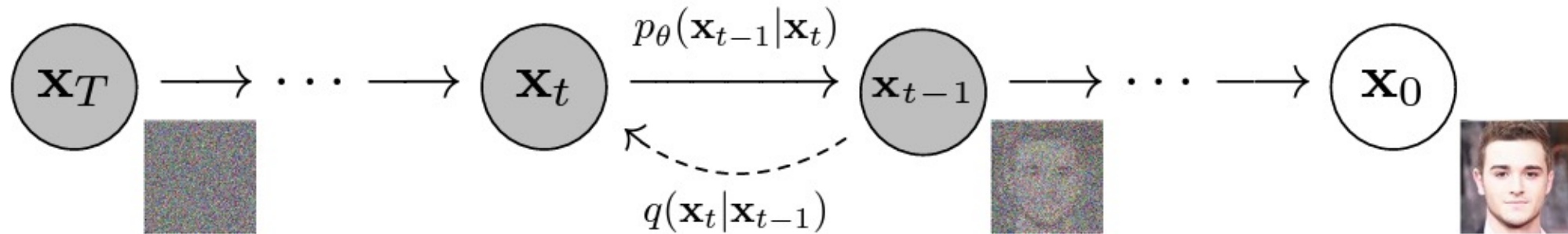
Agenda



Which are real? Which are fake?

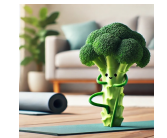
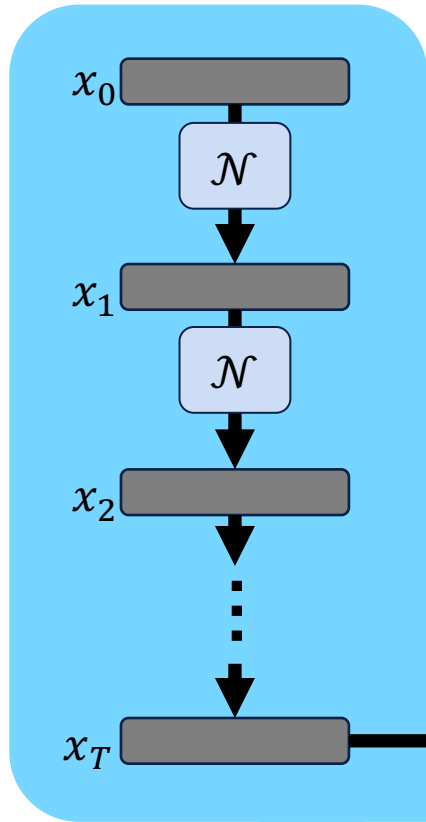
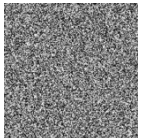


Denoising diffusion probabilistic models (DDPM) (Ho et al, 2020)

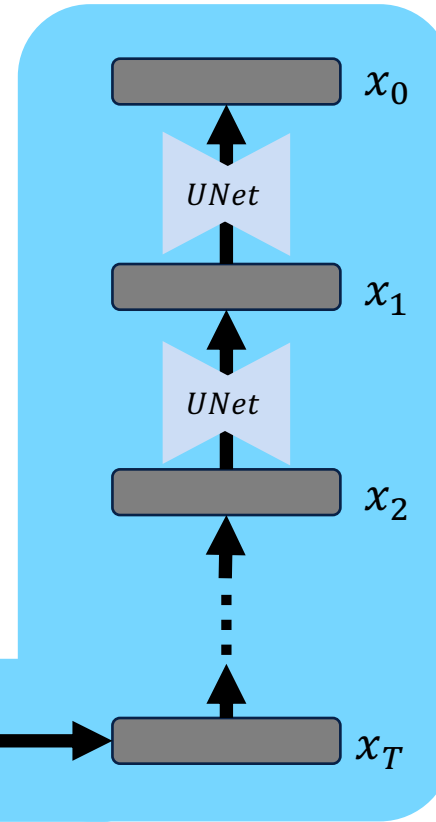
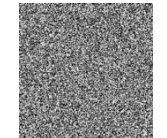




⋮



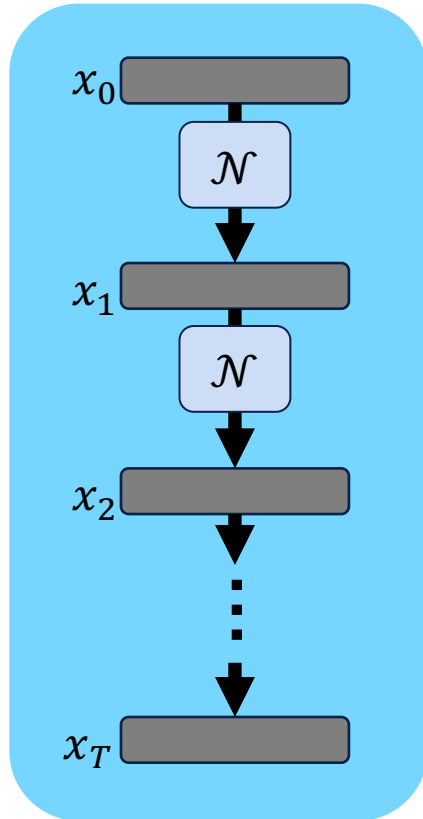
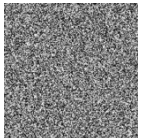
⋮



Forward process



⋮



For $t \leq T$, choose $\beta_t > 0$, let $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{s \leq t} \alpha_s$. Then,

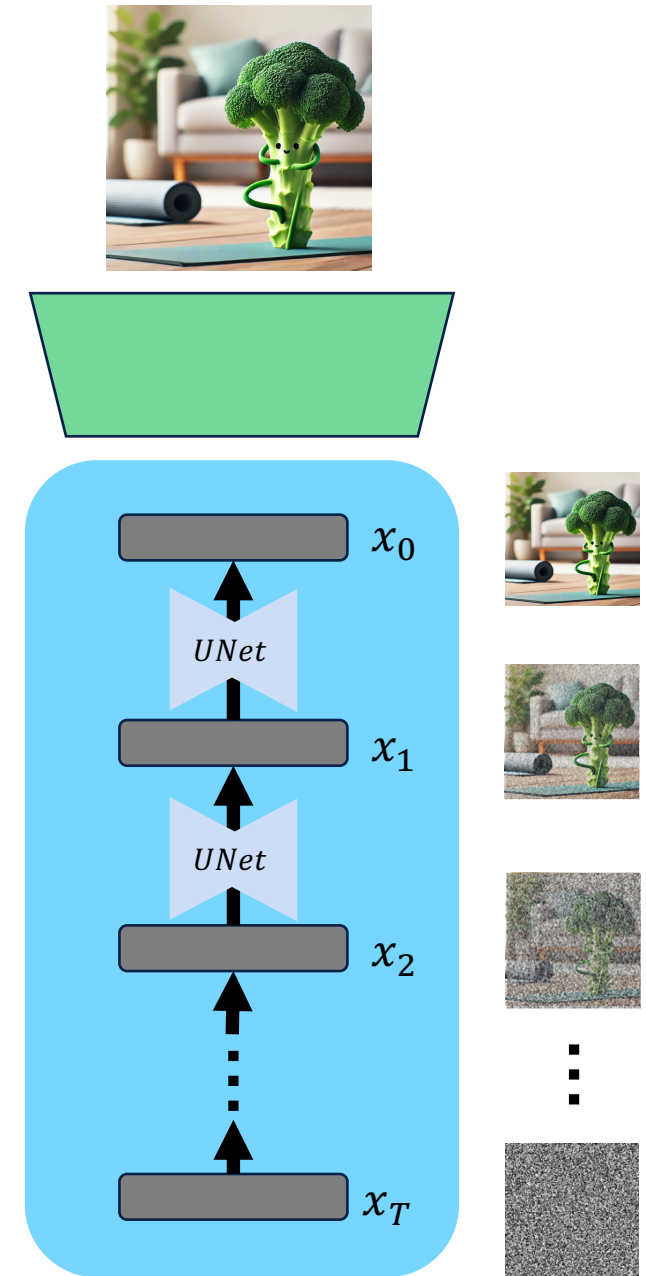
$$q(x_t | x_{t-1}) = \mathcal{N}(x_t | \sqrt{1 - \beta_t} x_{t-1}, \beta_t I)$$
$$q(x_t | x_0) = \mathcal{N}(x_t | \sqrt{\bar{\alpha}_t} x_0, (1 - \bar{\alpha}_t) I)$$

Reverse process

- For each $t \leq T$, the UNet tries to reconstruct **all noise** $\epsilon_\theta(x_t, t)$ that was added to form x_t from x_0 .
- We then try to reconstruct x_{t-1} as follows

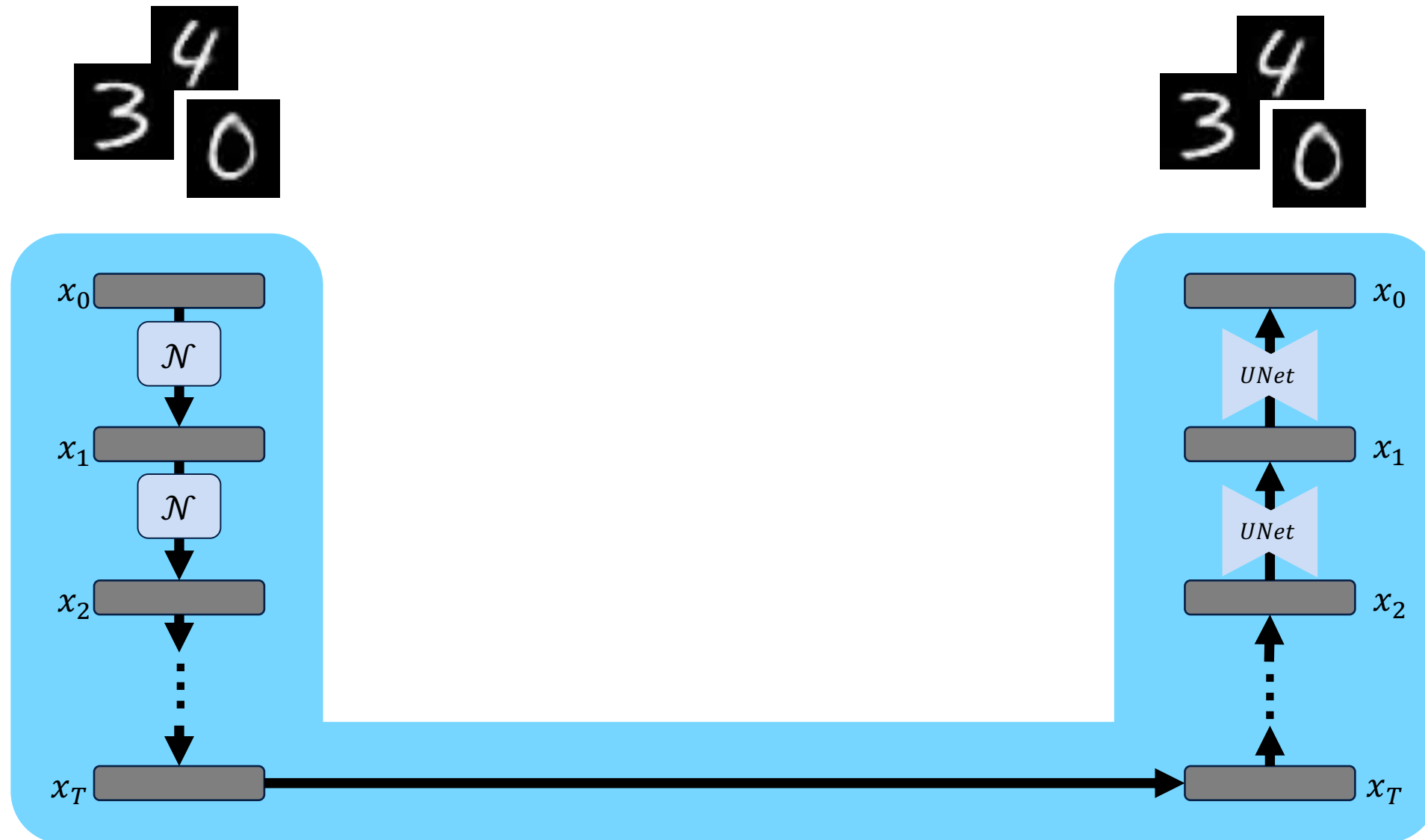
$$x_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{1-\alpha_t}{\sqrt{1-\alpha_t}} \epsilon_\theta(x_t, t) \right) + \sigma_t z,$$

where σ_t is a chosen hyperparameter and z is sampled from a standard Gaussian.

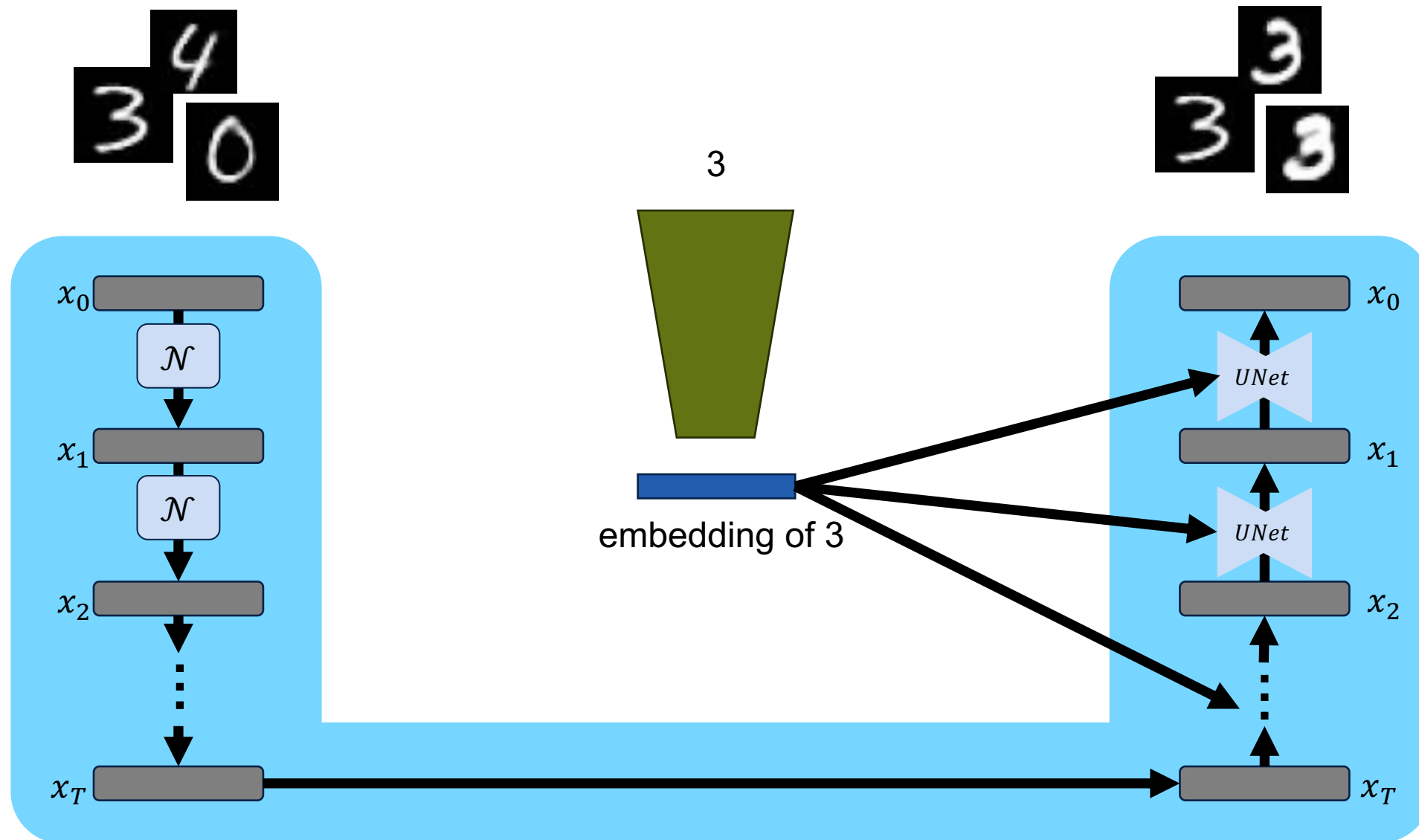


Conditional diffusion processes

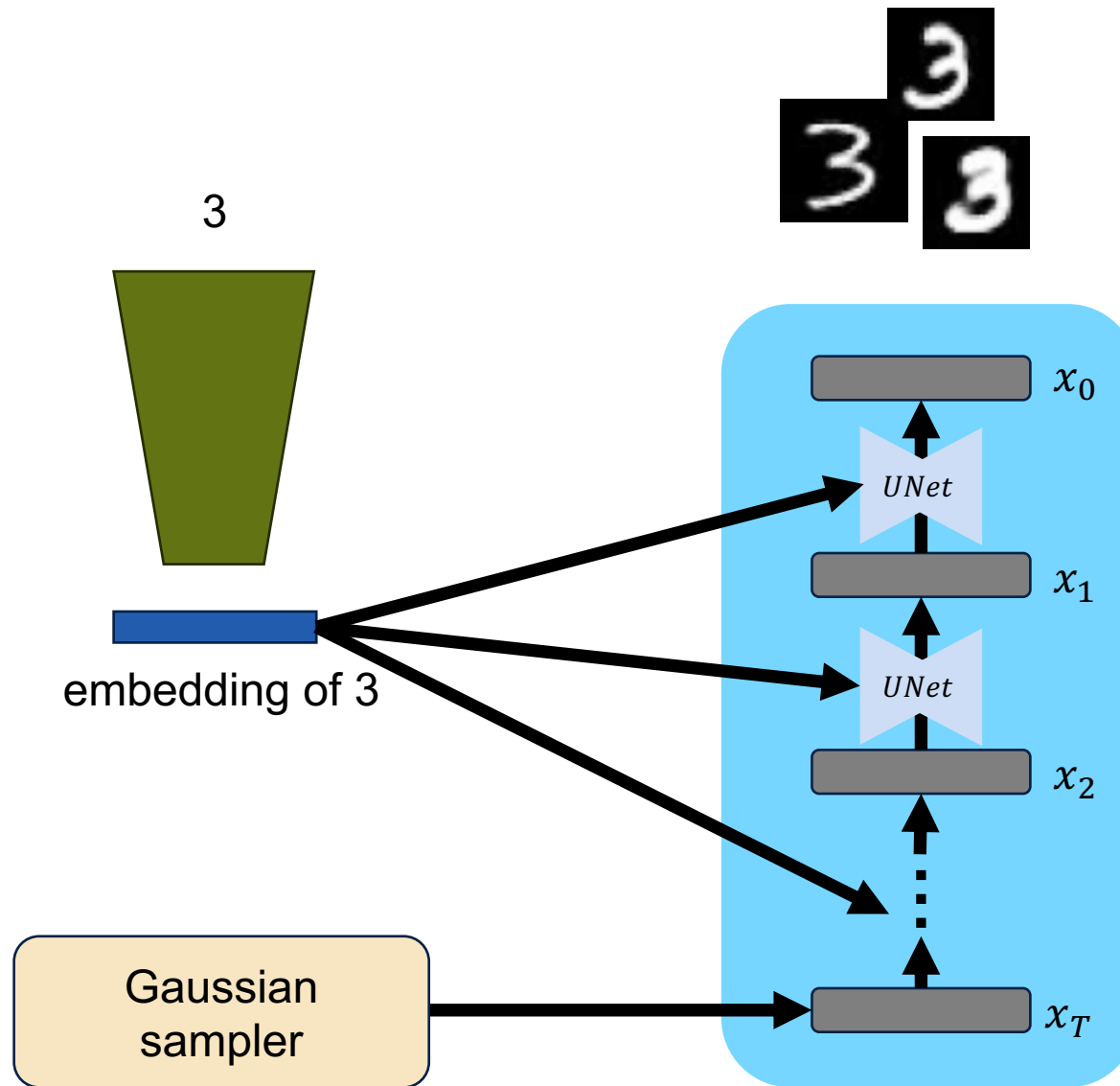
Diffusion process for MNIST



Conditional diffusion process for MNIST



Conditional generation of handwritten digits



Link to implementation with facehugger library

- <https://huggingface.co/learn/diffusion-course/en/unit2/3>